Singlet state mediated charge and spin dynamics in nitrogen-vacancy centers in diamond

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The unique optical properties of the nitrogen-vacancy (NV) center in diamond have established it as a workhorse for quantum science applications. Both charge and spin dynamics, driven by laser excitation, are critically mediated by the singlet state, yet several key mechanisms remain elusive. In this work, we provide experimental evidence of a novel charge conversion pathway, a transition from the neutral charge state (NV^0) to the singlet manifold of the negative charge state (NV^-) under high laser power. Furthermore, we directly confirm the two-photon ionization nature of the singlet state using near-infrared excitation and present an extensive discussion of the ionization mechanisms of NV^-. These findings offer deeper insights into the singlet-mediated charge-spin interplay, providing a foundation for potential advancements in spin readout technologies of NV centers.

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I. INTRODUCTION

The nitrogen-vacancy (NV) center in diamond has become one of the most versatile and reliable quantum systems in recent years [1–6], despite facing competition from various emerging quantum defects [7–12]. Its unique spin properties, featuring long coherence time and easy optical addressability at room temperature, enable sensitive detection of quantum states [13–16]. The charge-state properties, complementing the spin state, have fostered various quantum technologies, such as spin-to-charge conversion readout [17–19], optical nanoscopy [20,21], and photoelectrical spin readout [22,23], expanding the toolkit for quantum applications. However, unwanted stochastic charge-state conversion is a major source of decoherence, undermining spin initialization and readout performance. Therefore, a comprehensive understanding of charge-spin interplay is crucial for enhancing sensitivity and fully exploiting the potential of the NV center.

Although numerous studies have nearly completed the puzzle, some aspects of the underlying charge and spin dynamics remain unclear. In particular, the role of the singlet state in mediating the charge conversion has not been specifically explored. Recent research showed that the purity of a negatively charged NV center (NV^-) can be significantly enhanced by high optical power, leading to an increase in spin readout contrast [24]. With regard to what is responsible for this increase, it has been suggested that a strengthened recombination pathway from the neutral charge state (NV^0) to the singlet state of NV^- is involved. However, this argument has not yet been directly verified through experiments involving measurements of charge or spin states. On the other hand, the photoionization of NV^- is an indispensable factor during the process. It is widely accepted that ionization is a two-photon process, but the mechanisms—particularly when we compare ionization from the singlet state and the excited state—remain open to debate in the literature [25]. Clearly, the singlet state plays a critical role in the bidirectional charge conversion [26,27]. In this work, we focus on exploring the charge-spin interplay by providing direct experimental evidence of a novel singlet-mediated charge conversion pathway in which NV^0 transitions to the singlet manifold of NV^- under high laser power. In the process, tailored measurement schemes and careful data analysis are employed. Furthermore, we perform direct measurements to confirm the two-photon ionization nature of the singlet state using near-infrared excitation and present an extensive discussion of the ionization mechanisms of NV^-. Our results contribute to a clearer understanding of the complex mechanisms of charge and spin dynamics in the NV center.

A sketch of the NV's charge and spin structure is shown in Fig. 1(a). For the NV^-, green optical pumping induces either radiative emission back to the ground state without changing the spin states or spin-dependent cycling through nonradiative intersystem crossing (ISC) via the singlet state, in which the spin is not conserved. Repeated cycling through these probabilistic processes leads to spin polarization in the $m_s = 0$ state due to its higher probability of being the final state. During the cycling process, the $m_s = \pm 1$ spin states take more time, delayed primarily by the ∼200-ns shelving on the singlet ground state ^1E, and thus release less photoluminescence (PL) within the first few hundred nanoseconds after the laser is turned on. This differential emission forms the basis for distinguishing between spin states during spin readout.

To drive the inner spin dynamics of NV^-, a green laser (532 nm) is usually used. However, when the spin is pumped into the excited state or subsequently to the singlet state by one photon, the charge state of the NV center can be changed if a second photon is absorbed; this two-photon process is responsible for the ionization of NV^-. Simultaneously, the green laser also facilitates the conversion of NV^0 back to NV^- through a two-photon process: The center is first excited, followed by the capture of an additional electron from the va-

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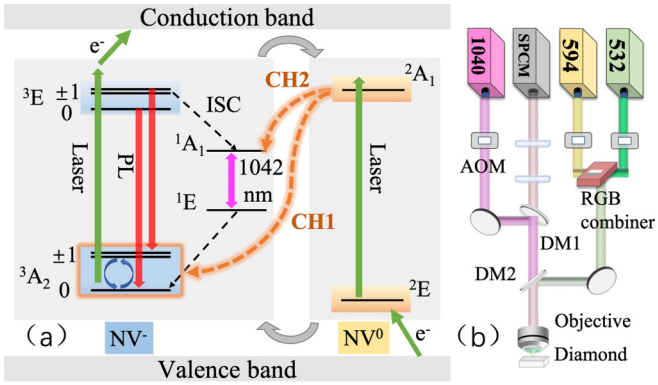


FIG. 1. (a) Diagram of energy levels and the charge-state conversion process of the NV center. The green arrows denote the 532-nm laser. The dashed yellow curves with arrows denote the two recombination pathways. (b) Experimental setup. The 532- and 594-nm lasers are first put together into one path with an RGB combiner and then combined with the 1040-nm laser at DM2. DM1 separates the returned fluorescence into a single-photon counting module (SPCM). DM1 and DM2 are long-pass dichroic mirrors with edge wavelengths of 950 and 650 nm, respectively. Acousto-optic modulators (AOM) are used for the switch.

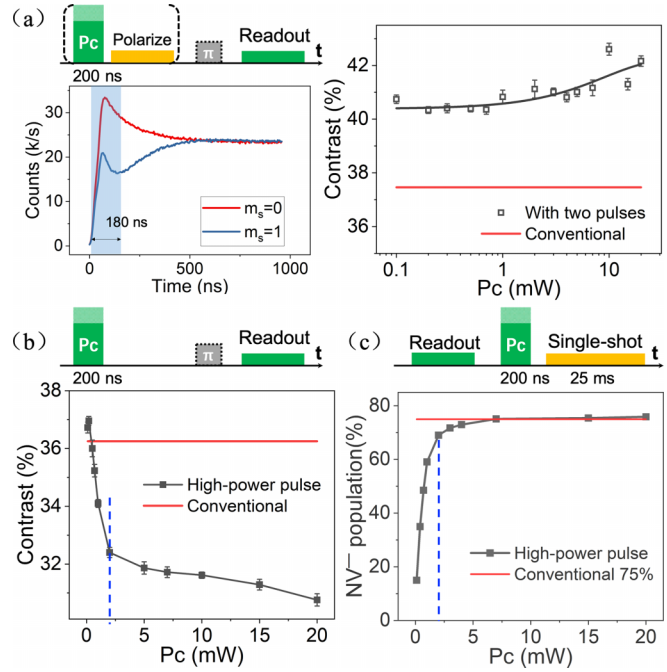


FIG. 2. (a) Spin contrast measurements with two-pulse initialization. The left panel shows the experimental pulse sequence and the PL collection window for the time traces. The 594-nm polarization pulse is set to 40 μW and 45 μs. The green readout pulse is 600 μW. The π pulse with the dashed outline indicates the presence or absence of a microwave π pulse for spin flipping. The solid black fitting line is for to guide the eye. (b) Spin contrasts measured without the polarization pulse. The solid red line denotes the averaged contrast over 10 sets of time traces measured in the conventional way. (c) Charge-state measurements after the P_c pulse. The yellow pulse for single-shot readout is set to 6 μW and 25 ms. The solid red line denotes the averaged NV^- population measured without the P_c pulse.

lence band. This recombination process is generally stronger than the ionization under green laser illumination, leading to a dynamic equilibrium where the NV^- population reaches around 75% [28]. Reference [24] suggested that two processes are responsible for the recombination: a normal direct process that drives the system directly to the NV^- ground state (CH1) and a novel Auger process that drives it to the singlet state (CH2). Evidence supporting the Auger process includes the fact that it is expected to be significantly faster than the direct process [29], and the energy required to convert the NV^0 ground state to the singlet state $^2A_1$ (4.3 eV) is twice the zero-phonon-line (ZPL) energy of NV^0 (575 nm, 2.156 eV). At high laser power, the model in [24] predicts that the NV^- population can be increased, even reaching greater than 92% in the limit. This prediction was supported by the observation of an enhanced PL contrast of the spins, achieved by applying a low-power polarization pulse following a high-power laser pulse. However, this experimental scheme was able to indicate only that the NV^- population was increased by the strong laser, without confirming that the increase originated from an enhanced population in the singlet state. Therefore, direct experimental evidence corroborating the singlet population is still necessary.

II. EXPERIMENTAL MEASUREMENTS OF CHANNEL CH2

To confirm the results, we first reproduced the two-pulse experiment. As shown in Fig. 2(a), we improved the pulse sequence by replacing the second green pulse for spin polarization with a weak yellow (594-nm) pulse, as this wavelength does not excite NV^0 [30]. The power and duration of this pulse were carefully adjusted to avoid any possible ionization. PL was collected as time traces for spins polarized in $m_s = 0$ and $m_s = 1$, respectively. The contrast was defined as $C = (C_0 - C_1)/C_0$ and calculated using a 180-ns time window for the maximum value, where C_0 and C_1 are the total photon counts for the two initial spins. As revealed by the experimental result, the spin contrast increases with the power of the first green pulse P_c, which prepares the charge state. Here, the duration of P_c was set to 200 ns, comparable to the singlet lifetime, to minimize its influence on the subsequent spin cycling. The large deviations observed at P_c powers above 10 mW are likely due to heat-induced diamond drifting in immersion oil, which can cause unstable PL over time. Although the contrast growth is small with increasing P_c power, the overall contrast induced by the two-pulse sequence is approximately 3% higher than in the approach without the two-pulse initialization (hereafter referred to as the "conventional" method), in which the pulse sequence consists solely of a square readout laser pulse. This is consistent with the previous result, verifying the enhancement effect of the two-pulse scheme on spin contrast [24,31].

Essentially, this enhancement is attributed to the high-power-induced increase in the NV^- population, which is then polarized by the second pulse. The interesting aspect of this sequence is that the two pulses operate the system in entirely different dynamic regimes: a high-power regime where com-

plex spin transitions between the two charge states compete in balance and a low-power regime where the dynamics occur in an almost isolated NV^- system. As a comparison, the result in the absence of the polarization pulse is shown in Fig. 2(b). Despite a slight increase when the P_c power is low (around 0.2 mW), the contrast decreases noticeably as the P_c power increases, leveling off when the power goes above 2 mW. This depolarization effect under high-power excitation was first proposed by Chapman and Plakhotnik [31] and was later clearly demonstrated by Chen *et al.* [32]. The origin of the depolarization lies in the charge-state conversion driven by cycling between the two charge states, particularly the process from the NV^0 excited state to the NV^- ground state (CH1), which is not spin selective. Now at high laser powers, a novel transition channel from the NV^0 excited state to the singlet state (CH2) is proposed to explain the increase in the NV^- population. A direct way to corroborate this channel is to measure the NV^- population right after the strong P_c is applied.

Therefore, we performed the experiment described by the time sequence in Fig. 2(c), where the polarization laser pulse is removed, and another low-power, long-duration 594-nm laser is inserted after the P_c pulse to perform charge-state measurement (see the single-shot technique in Ref. [30]). To accurately characterize the charge population in the experiment in Fig. 2(a), the green readout pulse is retained and shown in the beginning of the sequence since the entire sequence is looped. Actually, this pulse does not appear to have any influence on the experimental results. The measured NV^- population exhibits a rapid enhancement with rising P_c power, approaching a maximum of around 75% above 2 mW, which is the same as the value obtained with the conventional method. The reason why the NV^- population increase was not directly observed could be twofold: First, the increase itself is small, as indicated in Fig. 2(a), and second, the PL collection efficiency and thus the single-shot fidelity are limited by our measurement setup, which could be improved with the use of a solid immersion lens [18]. However, when comparing the two graphs in Figs. 2(b) and 2(c), we can observe a synchronous rapid change with increasing P_c up to 2 mW (marked by dashed blue lines), one decreases in contrast and one increases in NV^- population, followed by their simultaneous leveling off beyond this point. This behavior strongly suggests that the high-power green laser indeed boosts the NV^- population through extensive cycling between the two charge states.

Now we explore an alternative method to evaluate how the high-power pulse affects the singlet population. A new pulse sequence is presented in Fig. 3(a), where a near-infrared (NIR; 1040-nm) light pulse with fixed power and duration (60 mW, 300 ns) is introduced after the P_c pulse to ionize the population out of the singlet state before the remaining NV^- population is measured. This approach assumes that the ionization out of the singlet state is positively correlated with the singlet population. The 1040-nm wavelength is chosen for its efficient ionization, with no impact on spins in the ground state despite the high power [33]. The delay between P_c and the NIR pulse is set to 20 ns—longer than the excited state lifetime (~10 ns)—to allow for sufficient ISC to the singlet state, but it is much shorter than the singlet

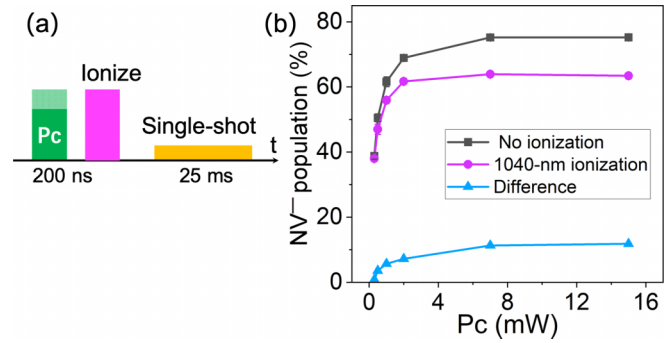


FIG. 3. (a) Experimental pulse sequence for ionization of the singlet state. A 1040-nm pulse (300 ns, 60 mW) is applied with a delay of 20 ns relative to the P_c pulse to achieve ionization. The charge state is finally measured with a single-shot pulse. (b) Experimental results of the sequence in (a). The light blue triangle curve is the difference of the other two curves.

lifetime to ensure that the ionization occurs before the population decays. As Fig. 3(b) shows, the NV^- population rises toward saturation with increasing P_c power and is reduced overall under NIR ionization. The difference curve (blue triangles) demonstrates the reduction, typically reaching about $11.81\% \pm 0.99\%$ when P_c is 15 mW. The rising trend of the difference curve clearly suggests that higher P_c power leads to a greater population in the singlet state, supporting the existence of channel CH2 for recombination.

III. DYNAMICS-BASED DATA ANALYSIS

However, the conclusion above is still tentative when two additional dynamical processes are considered. One is the depolarization through channel CH1 to the NV^- ground state, which increases the $m_s = \pm 1$ population. The other is the shelving process, which increases the singlet population via ISC from the excited state [17]. Consequently, the reduction $\Delta p_s = 11.8\% \pm 0.99\%$ in the NV^- population comprises contributions from two sources: channel CH2 and the combined effects of CH1 and internal spin transitions within NV^-. To evaluate the contribution from the latter source, we reanalyzed the data in Fig. 2(b) and performed backward calculations from the spin readout contrast with the initial populations, reconstructing the depolarized spin populations after the P_c pulse and just before the readout pulse for the time-trace measurement. This is achieved by modeling an isolated NV^- system using a five-level framework [Fig. 4(a)]. Following the method in Ref. [34], we first apply a nonlinear least-squares fit to the time-resolved fluorescence traces measured for spin states polarized in $m_s = 0$ and $m_s = 1$ and directly extract the transition rates between the ground, excited, and metastable states, as shown in Table I. These rates are then input into the rate equations to determine the relationship between the depolarized spin populations and PL contrast. Finally, the depolarized spin populations corresponding to the measured contrasts at different P_c powers are obtained using a one-dimensional data interpolation method, as shown in Fig. 4(b). During the time-trace fit, we use fixed excited state lifetimes of $t_0 = 1/(k_{35} + k_f) = 12$ ns and $t_{\pm 1} = 1/(k_{45} + k_f) = 8$ ns as boundary conditions, which are

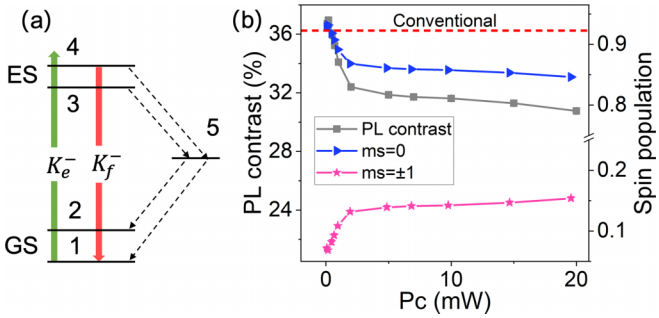


FIG. 4. (a) Five-level model used for evaluating spin populations from PL contrast. (b) Contrast data replotted from Fig. 2(b) and spin populations evaluated from the data. The spin population for $m_s = \pm 1$ is calculated by $p_1 = 1 - p_0$. The dashed red line marks the average contrast of 36.25% measured in the conventional way and the corresponding p_0 of 0.9217 on the two y axes.

extracted from a multiexponential fit of the decay following pulsed excitation [35,36]. In addition, the initial polarization rate is constrained to 0.92, which was measured with our experimental setup, consistent with the typical electron spin polarization with a square green laser pulse at room temperature [1,37].

The spin populations evaluated from the PL contrast demonstrate the depolarization results, where the population on $m_s = 0$ decreases to $p_0 = 0.8423 \pm 0.0024$ at 15 mW. These populations are normalized to the total NV^- population, satisfying $p_0 + p_1 = 1$. Since the time traces in Fig. 2(b) were measured long (~900 ns) after the P_c pulse was applied, p_0 and p_1 can be regarded as encompassing the whole population. p_1 primarily consists of the initial population 0.08 in $m_s = \pm 1$ state and the contribution from depolarization via channel CH1. p_0 includes the population remaining after depolarization and the population that decayed from the singlet state. Specifically, through the latter ISC process, p_0 additionally contains the majority of the population p_{CH2} that transitions through channel CH2 since $K_{51} > K_{52}$. A previous work [33] demonstrated that a 150-ns green pulse is sufficient to populate ~75% NV^- consistently; we therefore approximate NV^- as an independent system that does not lose spin population by the end of the P_c pulse. Now, we consider a scenario where a short, strong laser pulse excites the entire p_0 and p_1 population to the excited state without ionization. The maximum possible population in the singlet state is given by

$$p_s^{\max} = p_{NV^-}\left(p_1 \frac{K_{45}}{K_{45} + K_f} + p_0 \frac{K_{35}}{K_{35} + K_f}\right), \quad (1)$$

where K_{35}, K_{45}, and K_f are the respective transition rates and $p_{NV^-} = 75.24\% \pm 0.41\%$ is the measured spin population in NV^- when P_c is 15 mW. Therefore, in Fig. 3(a), before the NIR pulse is applied and while the dynamically populated singlet state remains almost intact, the singlet population arising from contributions other than CH2 cannot exceed $p_s^{\max}$. Inserting the transition rates and the values of p_0 and p_1 at 15-mW P_c into the equation, we calculate that $p_s^{\max} = 0.1291 \pm 0.0011$, which is even larger than the measured NV^- ionization $\Delta p_s = 0.1181 \pm 0.0099$. This result seems to suggest that no population originates through CH2. However, since Δp_s represents only a population change and may not account for the total population, it is premature to draw any definitive conclusion. To address this, we consider the equation $\dot{p}_s = -\gamma p_s$, which yields the result

$$p_s = p_{s0}\, e^{-\gamma t}, \quad (2)$$

where γ is the ionization rate of the NIR laser out of the singlet state and p_{s0} is the singlet population before ionization that we aim to determine. The following question then arises: How can the parameter γ be obtained?

Due to the limited lifetime of the singlet state, when γ is measured within this timescale, PL readout is expected to be more sensitive than the charge-state measurement. Accordingly, following the sequence in Fig. 3(a), we fix the P_c power at 15 mW and vary the duration of the NIR pulse from 50 to 370 ns. The single-shot 594-nm pulse is replaced with a shorter but stronger one to measure the fluorescence. The degree of ionization induced by the NIR pulse is determined based on the reduction in PL, normalized to the PL without ionization. The measured data are shown as black squares in Fig. 5(a), and fitting with the function $\epsilon(1 - e^{-\gamma t})$ yields $\gamma = 5.08 \pm 0.47$ MHz, where ϵ is an arbitrary coefficient. Given that the singlet population transitions to the ground state and then contributes to the PL, it is reasonable to suppose that this rate of change in PL closely approximates the rate of change in the spin population. Solving the equation $\Delta p_s = p_{s0} - p_{s0} e^{-\gamma t}$ gives $p_{s0} = 0.1510 \pm 0.0140$. Finally, the population that transitions through CH2 is evaluated as $p_{CH2} = p_{s0} - p_s^{\max} = 0.0219 \pm 0.014$. It should be pointed out that all the error values presented above were rigorously calculated in accordance with the law of propagation of uncertainties. Therefore, we hereby claim that, with the dynamics-based data analysis and correction, we have demonstrated experimental evidence of the existence of the proposed CH2 channel, through which the NV^0 charge state can directly transition to the singlet state of NV^- under high laser power.

However, the population p_{CH2} is very limited, accounting for only about 2% of the total spin population of an NV center. Such a small change aligns well with the fact that it was not detected in the charge-state measurement using our experimental setup, although it falls significantly below the prediction [24]. Considering the relatively large uncertainty, it is important to note that this value does not significantly

TABLE I. Spin transition rates used for numerical calculation in the five-level model. K_e is the laser excitation rate, and K_f is the emission rate. K_{ij} is the transition rate from state i to j.

Transition	Rate (MHz)
K_e	46.64
K_f	73.11
K_{35}	10.22
K_{45}	51.89
K_{51}	5.16
K_{52}	3.59

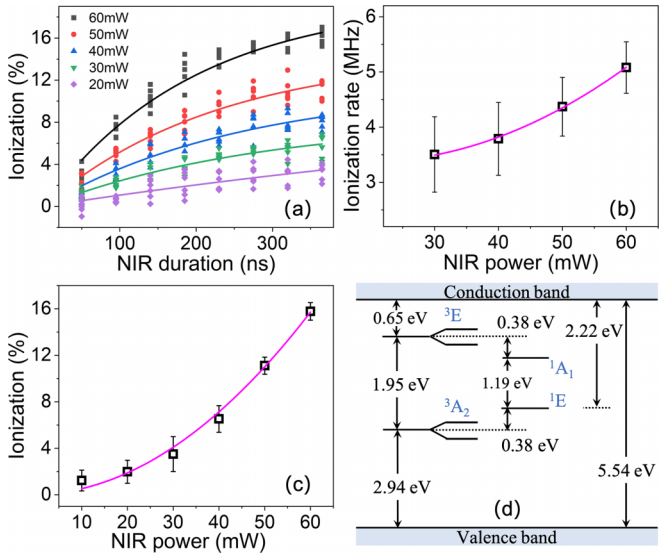


FIG. 5. (a) PL reduction due to NIR ionization from the singlet state. The pulse sequence is similar to that in Fig. 3(a), where the P_c pulse is fixed at 15 mW and the NIR pulse is varied over its duration from 50 to 370 ns at different powers. For each duration, the PL is measured six times with a 594-nm pulse of 80 μW for 500 ns, and the six values are all presented together. The degree of ionization is evaluated by PL reduction relative to the PL without a NIR pulse and is also normalized to the same value. The solid lines are fits to the function $\epsilon(1-e^{-\gamma t})$ according to Eq. (2), where ϵ is an arbitrary coefficient. (b) Ionization rate γ extracted from (a). The data are fitted using a quadratic function. (c) PL reduction due to NIR ionization of different powers. The NIR pulse duration is fixed at 200 ns, while its power is varied. The data are fitted with $\epsilon(1-e^{-\gamma t})$, where $\gamma=\alpha I+\beta I^2$, with I denoting the NIR power. (d) Diagram of energy levels of NV^-. The energy gaps between different levels are labeled according to the literature [1,28,38–40].

deviate from zero. The uncertainty primarily arises from the measurement error of γ, which is attributed to the small PL change relative to the substantial photon noise. Further improvements, such as employing higher-power pulses and implementing enhanced PL collection techniques, would be highly beneficial for reducing uncertainty and providing stronger evidence to confirm the role of CH2 with greater confidence.

IV. THE IONIZATION MECHANISMS OF NV^-

Evidently, the ionization of the singlet state using NIR light is not efficient, even at 60 mW. *Ab initio* calculations suggest that the ^1E level is about 0.38 eV above $^3A_2$ [41,42]. Considering the 1042-nm (1.19-eV) ZPL of the singlet state, it is reasonable to suggest that the ionization of the ^1E state to the conduction band (>2.22 eV) involves a two-photon process for NIR wavelengths. In such a process, high laser power is typically required to achieve sufficient ionization due to the small cross section of the singlet state [26,33]. While this two-photon process was proposed in Ref. [18], it has not yet been directly verified through experiments. To further investigate this mechanism, we extended the previous measurement of the ionization rate by measuring the PL reductions at more NIR powers, as shown in Fig. 5(a). The fitted ionization rates from these measurements are presented in Fig. 5(b), with the rate at 20 mW excluded due to its large deviation and fitting error. Even with this adjustment, the fitting for the rates exhibits a quadratic dependence on laser power that is too weak to definitively confirm the two-photon nature of the process. To improve this, we further refined our measurement scheme by fixing the NIR pulse duration while varying its power and directly measuring the corresponding PL reduction, as shown in Fig. 5(c). The data were fitted using the function $\epsilon(1-e^{-\gamma t})$, where $\gamma=\alpha I+\beta I^2$ is the ionization rate, with I denoting the NIR power and α and β being coefficients. This time, the fitting reveals a pronounced quadratic dependence of γ on the NIR power, with $\beta/\alpha>400$. Therefore, we experimentally confirm that the ionization from the singlet state using a NIR laser is a two-photon process.

In fact, NV^- can be ionized from both the excited state and the singlet state ^1E according to current research on and understanding of the issue, where $^1A_1$ is not considered due to its short lifetime (~1 ns) [26]. There has long been debate over which of the two ionization mechanisms dominates. While numerous studies have offered valuable insights from different perspectives, they have also led to a fragmented understanding. Here, we would like to provide a brief overview to clarify the issue based on representative papers. The 637-nm wavelength, as the ZPL of NV^-, is the one most commonly used for ionization and shows a strong preference for ionizing from the excited triplet state rather than the singlet [17,43,44]. Thus, it is typically employed in the spin readout scheme based on spin-to-charge conversion. The $m_s=\pm1$ population is first protected by shelving it in the singlet state; immediately after, the population in $m_s=0$ is ionized by 637-nm laser. It is important that the ionization occurs quickly, before the singlet state decays to the ground state. Two factors contribute to the ionization preference of 637 nm: On the one hand, the cross section of the singlet state is considered to be much smaller than that of the excited state [24]; on the other hand, a 637-nm photon does not have sufficient energy (1.95 eV) to excite the ^1E state to the conduction band.

From this perspective, a green photon (typically 532 nm, 2.33 eV) can excite both the triplet and singlet states. The question, then, is which state the green photon preferentially targets during ionization. Reference [25] showed that ionization of NV^- follows a quadratic dependence on the power of a 531-nm pulse (<100 ps) at low power and a linear dependence at high power when the excited state is saturated, confirming strong ionization from the excited state. Reference [45] demonstrates that increasing the delay between successive short 532-nm pulses (4 ns) within an initialization train from near zero to hundreds of nanoseconds enhances spin polarization, indicating weaker ionization from the singlet compared to the excited state. Furthermore, Ref. [46] directly proposed that 532-nm optical excitation cannot ionize NV^- from the shelving state, which actually requires higher photon energy. This is consistent with the absence of a positive contrast in photocurrent magnetic resonance detection [22].

For lasers with wavelengths longer than 637 nm, which have even lower photon energies, ionization is expected to occur preferentially from the excited state rather than the

singlet state. However, experimental results sometimes contradict this expectation. In Ref. [21], the authors demonstrated that a 780-nm pulse accelerates ionization only when applied in close temporal proximity to the NV center being pumped to the excited state. This finding confirms that a 780-nm laser can induce ionization from the excited state but does not rule out ionization from the singlet state, as the NV center is reinitialized to $m_s = 0$ in each sequence loop of their experiment. Reference [25], employing a finely constructed ultrashort-pulse sequence and modeling, demonstrated that a 766-nm pulse exhibits a slight preference for ionization from the singlet state, suggesting that ionization from the excited state is suppressed due to competition from stimulated emission. For NIR ($\sim$1000 nm) lasers, the preference for ionization from the singlet state is even more evident: When spins are initialized in $m_s = \pm 1$ and subsequently shelved in the singlet, the NV^- population decreases significantly with a longer shelving pulse before ionization or a shorter delay of the NIR ionization pulse after shelving [18]. The reason could be that the singlet state has a larger absorption cross section for NIR light than the excited state and the potential competition from stimulated emission around $\sim$1000 nm is negligible due to the short-lived $^1A_1$ state.

Generally speaking, unlike visible wavelengths (e.g., 532 and 637 nm) that primarily ionize the triplet state, NIR light preferentially targets the singlet state. Complementing previous studies, our results confirm the quadratic dependence of singlet ionization on NIR power. This wavelength-dependent ionization selectivity seems to imply that ionization from the triplet and singlet states could lead to the NV center reaching different final states of NV^0, which have been proposed to be the metastable ($^4A_2$) and ground (^2E) states, respectively [41]. Conversely, we imagine that the special transition from NV^0 to the singlet state of NV^- occurs because, under strong green light, NV^0 is more likely to occupy an initial spin state that favors the singlet. As a result, the mechanisms outlined above form the basis for on-demand charge-state control. Since the state-of-the-art charge initialization techniques rely on real-time feedback for NV^- preparation, improved NV^- initialization fidelity could speed up experiments [37,47]. On the other hand, the probability of shelving $m_s = \pm 1$ from the excited state in the singlet state is $\sim$42% (calculated according to Table I, consistent with previous studies [48]), implying that the current spin-to-charge readout of $m_s = 0$ contains about half of the $m_s = \pm 1$ spin. However, since only a small fraction of $m_s = 0$ spins ($\sim$12%) cross over into the singlet, the spin-to-charge readout will predominantly reflect the $m_s = \pm 1$ spin population when a singlet ionization scheme using strong NIR light is applied, despite only around 42% of the spins being read out. Repeating the shelving-ionization process two or more times could further improve readout fidelity. Therefore, our study highlights the critical role of the singlet state in the charge conversion process, opening promising avenues for advancements in NV center spin readout technologies.

V. CONCLUSION

In summary, we experimentally investigated the charge-spin dynamics mediated by the singlet state in the NV center in diamond. Through tailored measurement schemes and dynamics-based data analysis, we provided direct evidence of the existence of a novel charge conversion channel from NV^0 to the singlet state under high laser power. Further measurements of fluorescence reduction upon NIR-induced ionization from the singlet revealed a quadratic dependence on NIR power, confirming the two-photon nature of this process. An extensive discussion of the ionization mechanisms of NV^- was also presented, comparing ionization from the singlet and excited states under various excitation wavelengths. Although the measured population transferred through channel CH2 is small and limited by our experimental precision, our results offer valuable insights into the complex mechanisms underlying charge-spin transitions. Improvements in experimental techniques—such as higher-power lasers and enhanced PL collection—will be essential for reducing uncertainties and enabling further validation of these mechanisms with greater confidence. Our findings provide a foundation for potential improvements in spin-to-charge conversion and spin readout fidelity for NV center based quantum technologies.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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